

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Functional Features Overview

The Emotion Detection & Learning Support Engine provides real-time emotion detection from user text and delivers personalized learning support through AI-powered recommendations. The system includes text preprocessing, BERT-based emotion classification, an interactive chatbot, and a user-friendly Streamlit interface.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Input Form	Accepts user text for emotion analysis	Streamlit Frontend	Done	High
2	Text Preprocessing	Cleans and prepares input text for prediction	Preprocessing Module	Done	High
3	Emotion Detection	Predicts emotions using the fine-tuned BERT model	Emotion Detection Module	Done	Medium
4	Personalized Learning Recommendations	Suggests suitable learning resources based on detected emotions	Recommendation Module	Done	High
5	AI Chatbot Support	Provides personalized guidance using the Gemini API	Chatbot Module	Done	High
6	Emotion Result Display	Displays predicted emotion with confidence score	Prediction Module	Done	Medium
7	Interactive User Interface	Provides an easy-to-use Streamlit web interface	Frontend	Done	Medium
8	Error Handling	Handles invalid inputs and API/model exceptions	Backend	Done	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Text Input, Emotion Result Display
2	Backend / Logic	3	Emotion Detection, Recommendation Engine, Error Handling
3	Model / Storage	1	Fine-tuned BERT Model and Tokenizer
4	API / Integration	1	Google Gemini API Integration
5	Security / Authentication	0	Not Implemented